GUIDE
Building a vSphere 8 Nested Lab on AMD Ryzen 6000

v1.1.0

Author:

Eldon Gabriel

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Disclaimer: This report documents my independent laboratory research and configuration work related to enabling nested virtualization on an AMD-based Windows 11 system using VMware Workstation. The VMware ESXi deployment phase was not completed because of licensing restrictions requiring a corporate account for evaluation access.

This documentation reflects the author’s personal testing, troubleshooting, and validation steps performed in a controlled lab environment. No MCSI video content, proprietary lab materials, or protected instructional resources have been shared or distributed. All explanations are written in the author’s own words and follow MCSI’s disclosure and academic integrity policies.

REVISION HISTORY

Version	Date	Author	Description of Changes
1.0.0	02/22/2026	Eldon G.	Initial draft.
1.1.0	03/10/2026	Eldon G.	Major revisions and technical validation. Expanded firmware configuration section. Added OS-level virtualization conflict resolution. Added recovery procedures and troubleshooting workflow. Added security tradeoff analysis and framework alignment.

1.0 AMD NESTED VIRTUALIZATION

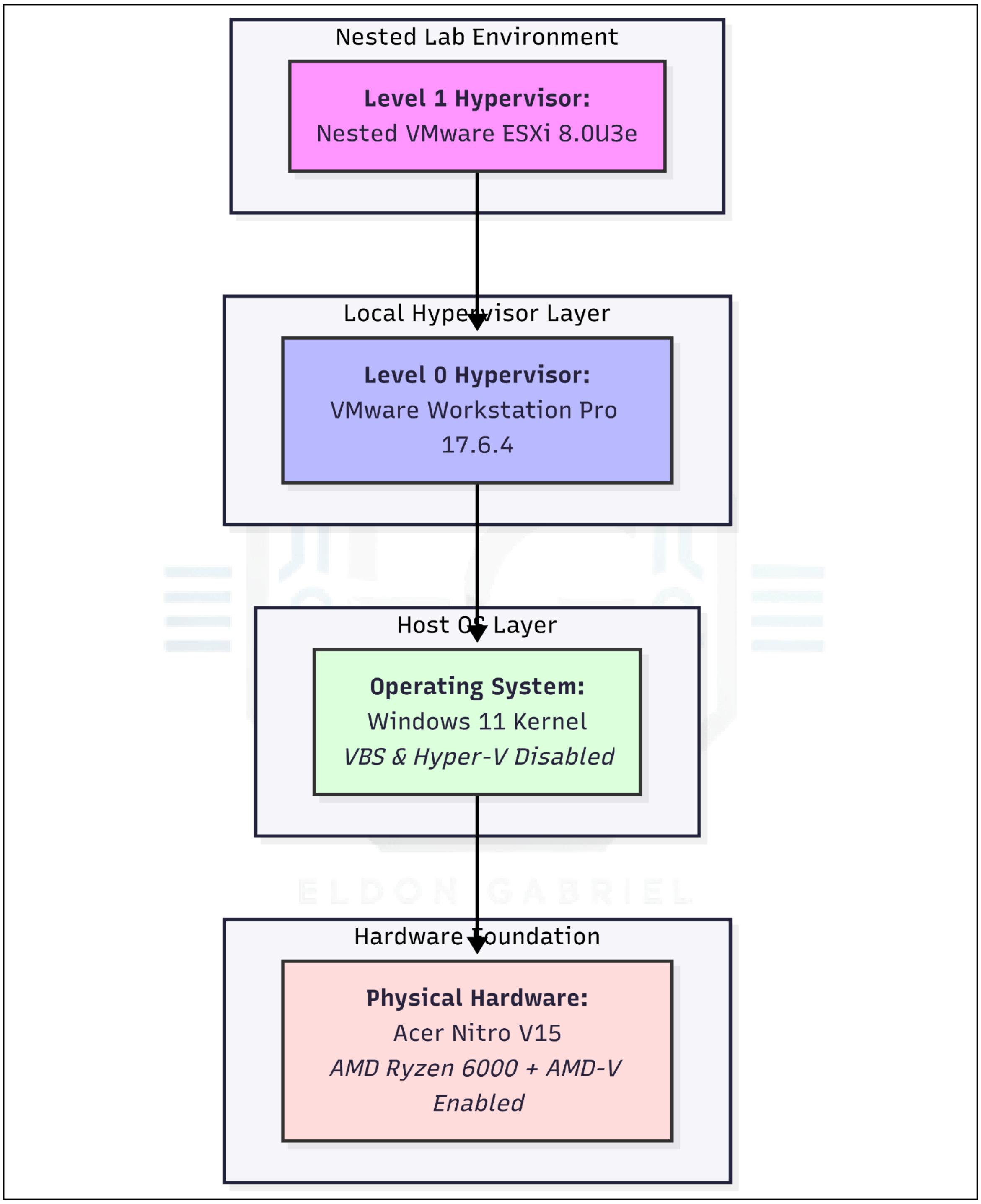

Figure 1: Nested virtualization stack on AMD Ryzen 6000. February 22, 2026. Eldon G.

1.1 Project Description

This guide explains how **nested virtualization** was enabled on the **AMD Ryzen 6000 laptop platform**. Initially, the **VMware Workstation** could not detect **AMD-V (SVM),** even though the CPU supported hardware virtualization.

The objective was to deploy a **VMware vSphere 8.0U3e lab environment** using **VMware Workstation Pro 17.6.4**.

The laboratory environment comprised the following:

- Nested **VMware ESXi hosts**
- **vCenter Server Appliance (VCSA)**

Lab Environment

Component	Specification	Lab Role / Allocation
System	Acer Nitro V15 (ANV15-41)	Host platform
CPU	AMD Ryzen 5 6600H (6C/12T)	Provides SVM/RVI hardware extensions
RAM	32GB DDR5	Supports Host OS + Nested ESXi + VCSA (Tiny deployment)
Storage	512GB NVMe Gen4	Datastore for nested VMs and vCenter
Host OS	Windows 11 Home	Management Layer (VBS/Hyper-V disabled)

During deployment, several technical issues had to be resolved:

- Hidden **firmware virtualization settings**
- **Windows 11 Hyper-V and VBS conflicts**
- **Incorrect chipset drivers installed by Windows Update**
- Windows security protections **blocking hypervisor access to AMD-V**

All security configurations were restored after lab validation.

1.2 Security and Risk Disclaimer

This procedure requires modifying the **firmware-level configuration** and disabling certain **Windows security protections**, such as

- **Virtualization-Based Security (VBS)**
- **Hyper-V**
- **Local Security Authority (LSA) Isolation**

Disabling these protections reduces the defense-in-depth posture of the system.

These changes were made **only in a controlled laboratory environment** for virtualization testing.

In production environments, these settings must comply with **organizational security policies** and be **restored after testing**.

1.3 Chipset and Driver Synchronization (AMD + Windows Alignment)

For nested virtualization to function correctly, **Windows must detect the SVM virtualization state provided by the firmware**.

During testing, the **default drivers installed by Windows Update** did not correctly expose virtualization capabilities to the VMware Workstation. This resulted in **inconsistent AMD-V detection**.

Action

The **latest AMD Chipset Software** is installed directly from the AMD.

Objective

Ensure proper communication between:

- System firmware
- AMD chipset controller
- Windows kernel virtualization layer

Critical Drivers for Virtualization

AMD PSP Driver: Ensures that Windows correctly interprets **firmware security states** through the **Platform Security Processor (PSP)**.

AMD PMF Driver: Improves coordination between the firmware and operating system during **power management and virtualization workloads**.

These components help ensure that the OS properly exposes **AMD-V (SVM)** capabilities to the hypervisor.

Windows Feature Alignment

After updating the chipset drivers, the Windows components required for virtualization feature detection were enabled.

Navigate to:

Control Panel → Programs → Turn Windows features on or off

Enable:

- **Virtual Machine Platform**
- **Windows Hypervisor Platform**

These components allow Windows to **register virtualization support appropriately within the kernel**.

Reboot Requirement

A **full system restart** is required after installing the chipset drivers and enabling Windows features.

*Note: In certain **Windows 11 updates**, these virtualization features can sometimes **re-enable portions of the Windows hypervisor**, even if **VBS has been disabled through registry configuration**. This behavior should be validated in later verification steps.*

1.4 Firmware Initialization: Hidden UEFI Access

The **AMD Ryzen 6000 series platform** requires specific firmware-level configurations to support **Type-1 hypervisor nesting**. Many laptop manufacturers hide advanced CPU configuration menus, preventing direct access to virtualization controls, such as the **SVM Mode** and **System Management Mode (SMM) protections**.

Owing to this restriction, a **vendor-specific firmware unlock sequence** is required to expose the hidden **Advanced CPU configuration menus** within the UEFI interface.

Diagnostic Reasoning for Firmware Configuration

Layer Identification

This phase targets the ***Physical / Firmware Layer*** to ensure that the CPU exposes hardware virtualization instructions to the host operating system and the hypervisor stack.

Constraint

Standard BIOS/UEFI menus on consumer laptops often omit the **Advanced configuration tab**, which is required to modify CPU virtualization flags.

Validation

Successful access is confirmed once the **Advanced Firmware menu becomes visible**, allowing modification of the CPU-specific virtualization and security parameters.

Startup Holding Method (Firmware Unlock Sequence)

Some AMD Ryzen laptop platforms hide advanced firmware menus behind vendor-specific startup sequences. The following procedure was required to unlock the hidden configuration menu on **Acer Nitro V15 (Ryzen 6000 platform)**.

Stage 1 – System Shutdown

Complete shutdown of the laptop.

Stage 2 – Key Combination Initiation

Press and hold **Fn + Tab** simultaneously.

Stage 3 – Power-On Sequence

While holding both keys, press the **Power button**.

Stage 4 – Firmware Trigger

The keys are released once the backlight of the keyboard is **illuminated**.

Stage 5 – BIOS Entry

Immediately, press **F2 repeatedly** to enter the BIOS interface.

Stage 6 – Authentication

Enter the BIOS password when prompted *(see Figure 2)*.

Stage 7 – Advanced Menu Access

Once the menu becomes visible, navigate to the initial **Advanced** option *(see Figure 3)*.

Stage 8 – CPU Configuration Access

Navigate to the **CPU Related Settings** to review and modify virtualization-related firmware flags *(see Figure 4)*.

Required CPU Virtualization Firmware Configuration

Setting	Correct State	Reason
SVM Support	Enabled	Enables AMD-V virtualization extensions
SVM Lock	Disabled	Allows the hypervisor to manage SVM state dynamically
SMM Code Lock	Disabled	Prevents System Management Mode from blocking nested hypervisor calls
SMM Protection	Disabled	Prevents firmware security controls from intercepting virtualization instructions used by nested hypervisors
SMM Isolation Support	Enabled	Maintains basic firmware isolation safeguards while still allowing nested virtualization

After applying the required configuration, **F10** is pressed to save the updated **NVRAM state** and reboot the system.

Expected Result

The system successfully boots into **Windows 11** with hardware virtualization.

Verification can be performed using

Task Manager → Performance → CPU

Expected value:

Virtualization: Enabled

This confirms that the firmware correctly exposes the **AMD-V/SVM extensions**, allowing VMware Workstation to initialize nested virtualization features.

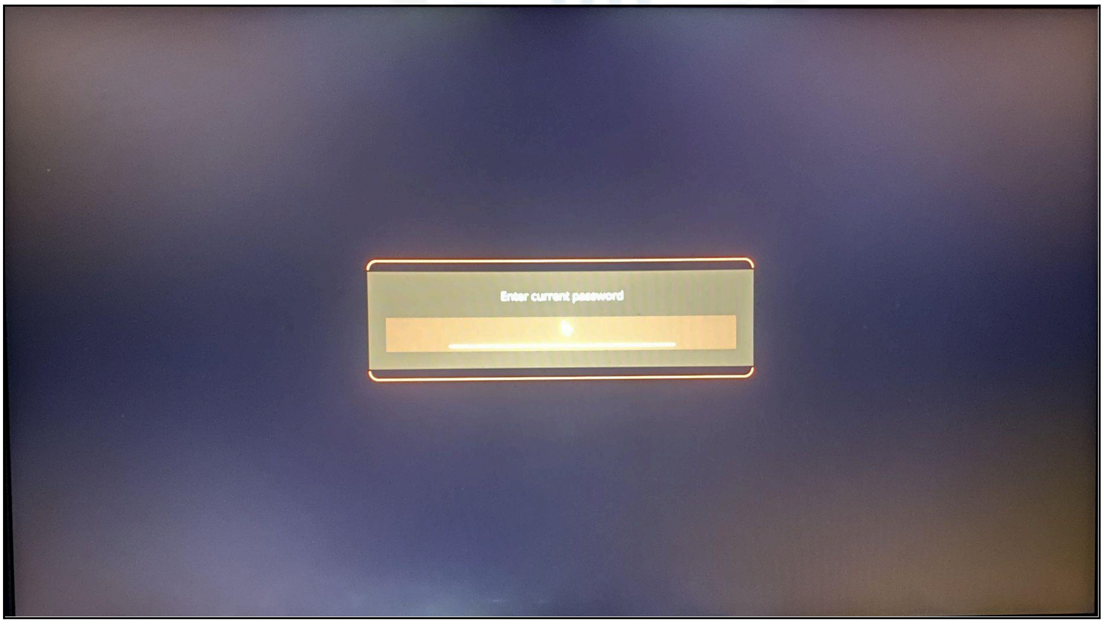

Figure 2: Acer Nitro V15 BIOS Password Screen. March 10, 2026. Eldon G.

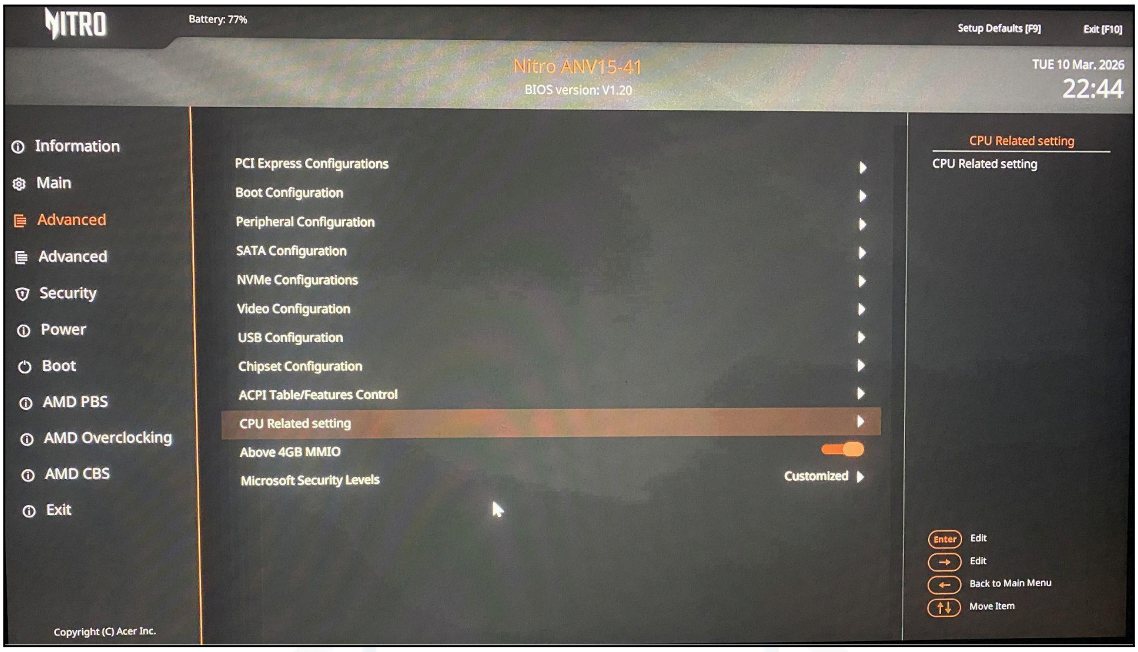

Figure 3: Acer Nitro V15 Advanced BIOS Interface. March 10, 2026. Eldon G.

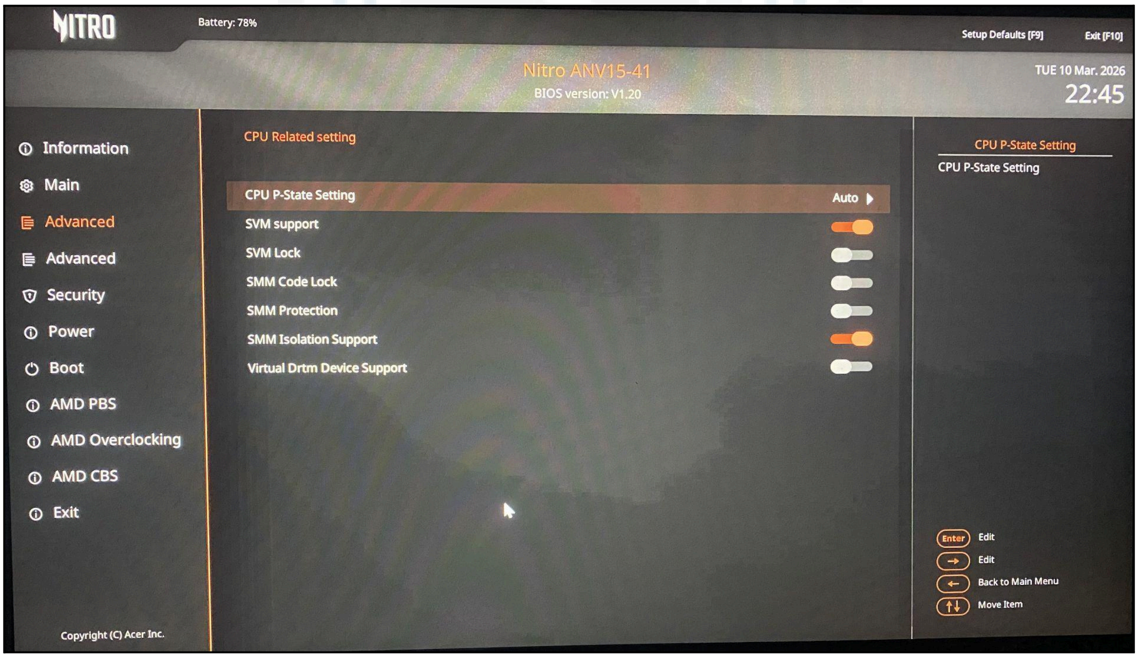

Figure 4: Acer Nitro V15 Advanced BIOS Interface. March 10, 2026. Eldon G.

1.5 OS-Level Configuration: Disabling VBS

Windows 11 uses hardware virtualization to power several built-in security features, such as

- **Virtualization-Based Security (VBS)**
- **Credential Guard**
- **Memory Integrity (Core Isolation)**

When these protections are enabled, **Windows loads its hypervisor layer**. This prevents the **VMware Workstation** from accessing **AMD-V (SVM)** directly.

To allow nested virtualization, these Windows security features must be **disabled so that VMware can obtain exclusive control of the virtualization extensions**.

PowerShell Configuration (Run as Administrator)

Execute the following commands from an **elevated PowerShell session**:

A **system reboot is required** after completion of these steps.

Disable the Windows Hypervisor

```powershell
bcdedit /set hypervisorlaunchtype off
```

This prevents Windows from loading the **hyper-V hypervisor at the system startup**.

Disable Virtualization-Based Security (VBS)

```powershell
reg add "HKLM\SYSTEM\CurrentControlSet\Control\DeviceGuard"
/v "EnableVirtualizationBasedSecurity" /t REG_DWORD /d 0 /f
```

This disables the VBS framework that is responsible for isolating sensitive kernel processes.

Disable Local Security Authority (LSA) Isolation

```powershell
reg add "HKLM\SYSTEM\CurrentControlSet\Control\LSA" /v "LsaCfgFlags"/t REG_DWORD /d 0 /f
```

This disables **LSA protection**, which otherwise runs in a protected virtualization container.

Credential Guard De-Provisioning (If UEFI Locked)

In some systems, **Credential Guard remains locked in the firmware, even after registry changes**.

If this occurs, run the Microsoft **Device Guard Readiness Tool**.

```powershell
.\DG_Readiness_Tool_v3.6.ps1 -Disable -Execute
```

During the next reboot, the system displays a **firmware confirmation prompt**.

Follow on-screen instructions to complete the **Credential Guard opt-out process**.

Verification Procedure

To confirm that the **VBS has been successfully disabled**, perform the following validation.

Step 1
Right-click the **Start** button.

Step 2
Select **Run**.

Step 3
Type: `msinfo32`

Click **OK**.

Step 4
Scroll to **Virtualization-based security**.

Step 5
Verify that the status is as follows:

Virtualization-based security: Not enabled

(See Figure 7)

If this step is skipped, Windows may continue loading its hypervisor, and **VMware will not be able to initialize nested virtualization**.

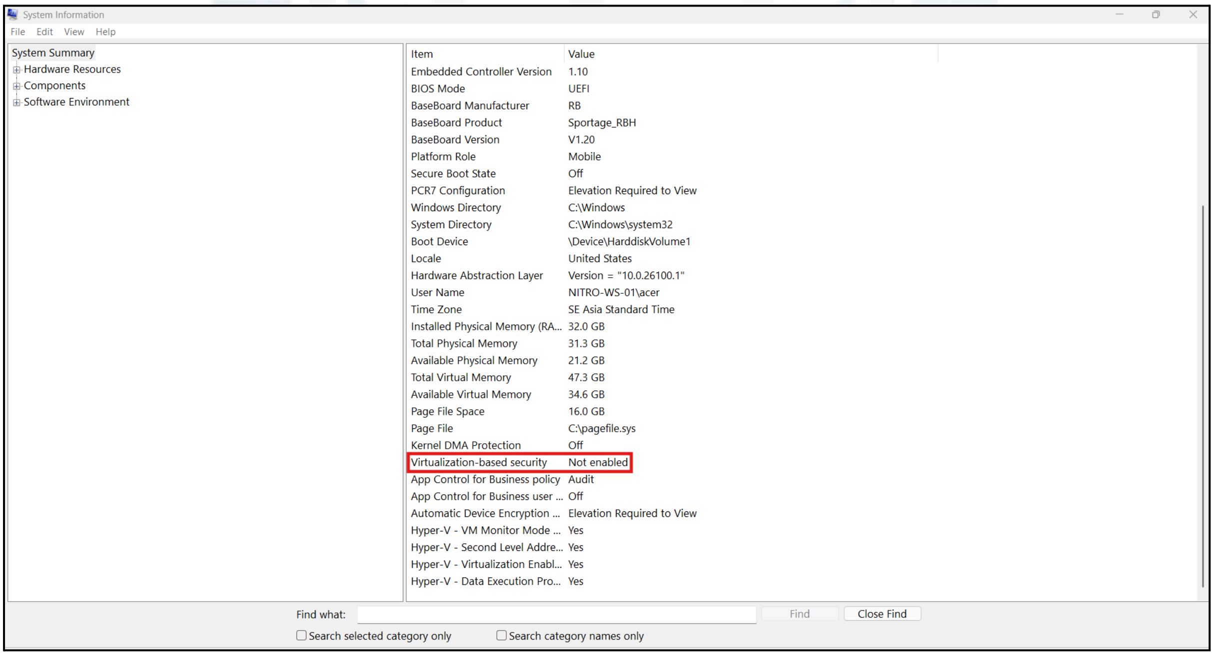

Figure 5: Windows System Information – VBS Status Validation. March 10, 2026. Eldon G.

1.6 Incident Recovery: CMOS / NVRAM Reset

An incorrect firmware configuration can, at times, place the system into a **no-Power-on Self-Test (no-POST) condition**.

1.6.1 Common Symptoms

The system powers on but fails to initialize the firmware interface. Typical indicators include:

- **Black screen**
- **Cooling fans running continuously**
- **Keyboard backlight active**
- No BIOS splash screen

This behavior typically indicates **conflicting firmware configuration values stored in NVRAM**, particularly when modifying advanced CPU security settings related to virtualization.

1.6.2 Recovery Procedure: Battery Pin-Hole Reset

Most modern laptops include a **hardware battery reset mechanism** that clears temporary firmware states and forces the system to reinitialize the hardware configuration.

Stage 1 – Disconnect Power

Ensure that the **AC power adapter** is disconnected from the laptop.

Stage 2 – Locate Reset Pin-Hole

Turn the laptop over and locate the **battery reset pin hole** on the underside of the chassis.

Stage 3 – Execute Reset

Using a **non-conductive pin or paperclip**, press and hold the internal reset button for **20 s.**

Expected Result

The reset clears **volatile NVRAM configuration states**, allowing the system firmware to reload **default parameters** during the next boot.

After the reset:

- The system should successfully **POST**
- BIOS/UEFI access should be restored
- Firmware settings may need to be **reconfigured for virtualization**

1.7 Validation and Verification

After completing the firmware configuration, chipset synchronization, and OS-level security changes, the system must be validated to confirm that **hardware virtualization is fully available to VMware Workstation**.

Validation was performed using **three independent checks**.

Step 1 – VBS Status Verification

Tool: `msinfo32`

Open the **Windows System Information** utility and verify the following fields:

Virtualization-based security → Not enabled

If VBS remains enabled, Windows will continue to load its internal hypervisor layer, preventing VMware Workstation from accessing **AMD-V (SVM)**.

Step 2 – Hypervisor Capability Verification

Tool: `systeminfo`

Execute the following command from a **Command Prompt or PowerShell session**:

```
systeminfo
```

Verify that the following results appear in the **Hyper-V Requirements** section:

Virtualization Enabled In Firmware → Yes

Second Level Address Translation → Yes

VM Monitor Mode Extensions → Yes

The output **must not display** the message:

A hypervisor is detected.

If this message appears, the **Windows Hypervisor is still active**, and the VMware Workstation will not be able to initialize nested virtualization.

Step 3 – VMware Nested Virtualization Verification

Open the **ESXi virtual machine settings** in the VMware Workstation.

Navigate to:

VM Settings → Processors

Select the following option:

- **Virtualize Intel VT-x/EPT or AMD-V/RVI**

(See Figure 7)

Expected Result

Successful validation confirms the following:

- **ESXi boots without AMD-V errors**
- VMware detects **hardware virtualization extensions**
- **Nested 64-bit virtual machines** can be created inside ESXi

This confirms that **nested virtualization is fully operational**.

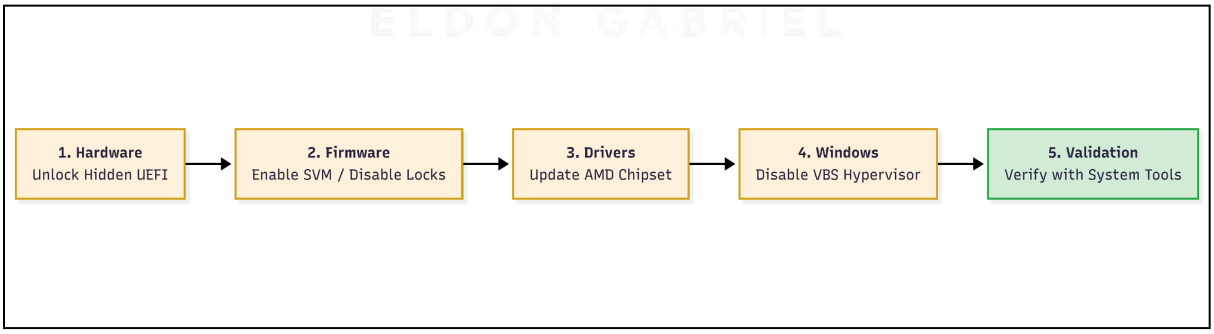

Figure 6: Engineering workflow for platform enablement. 2026. Eldon G.

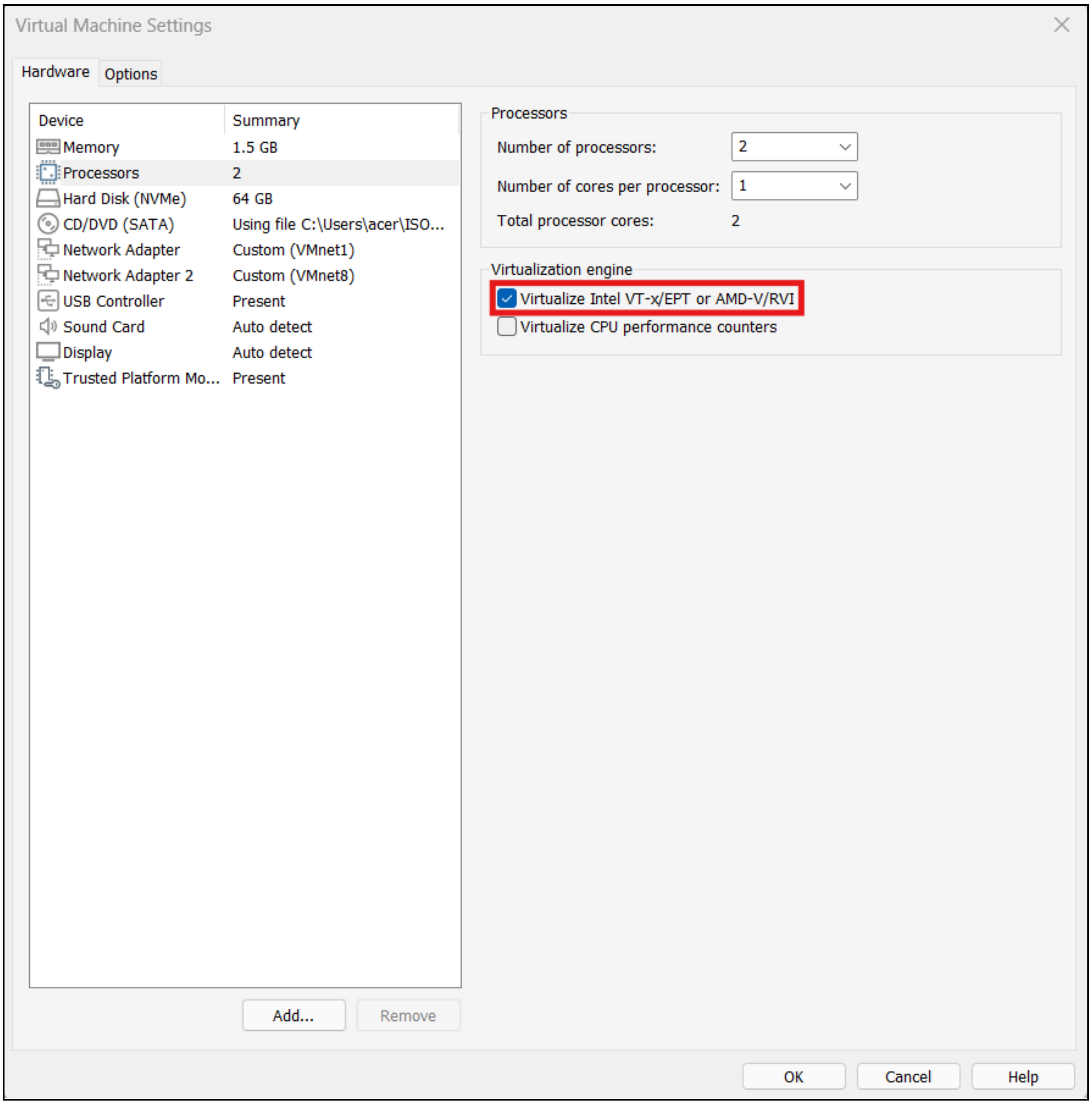

Figure 7: VMware virtual machine Settings. March 10, 2026. Eldon G.

1.8 Troubleshooting Decision Tree

The following table summarizes the most common issues encountered when enabling nested virtualization on **AMD Ryzen mobile platforms** and the corrective actions required to resolve them.

Symptom	Likely Cause	Resolution
AMD-V Not Supported	VBS still active or SVM disabled in firmware	Verify that **Virtualization-Based Security is disabled** using msinfo32. Confirm **SVM Mode is enabled** in BIOS/UEFI.
Black Screen / No POST	Conflicting firmware security settings (SMM configuration)	Perform a **battery pin-hole reset** to clear the NVRAM. Restore the **SMM settings to the default** and then reconfigure the virtualization options.
Hyper-V Still Detected	Windows hypervisor components still active	Disable **Memory Integrity (Core Isolation)** and remove all **Hyper-V-related Windows features**. Re-run the following commands and reboot: `bcdedit/set hypervisorlaunchtype off`
VCSA Installation Stalls at 80%	Infrastructure configuration problem or insufficient resources	Verify DNS resolution, time synchronization, and ensure sufficient RAM allocation for deployment.

1.8.1 Engineering Note: Hardware Resource Headroom

The standard **configuration of the Acer Nitro V15 shipped with 16GB of RAM**. For this lab environment, the system was upgraded to **32GB of DDR5** to provide sufficient capacity for nested virtualization workloads.

This configuration supports the following allocations:

Component	Typical Memory Allocation
Windows 11 Host OS	6–8 GB
Nested ESXi Host	4–8 GB
VCSA Tiny Deployment	12 GB

Providing an additional memory headroom prevents **host-level memory contention**, which can otherwise cause severe performance degradation, disk swapping, or **failed VCSA deployments**.

1.9 Security Tradeoff Summary

To enable nested virtualization, several **firmware and operating system security protections were temporarily disabled**. These changes allowed VMware Workstation to access **AMD-V (SVM) hardware extensions directly**, which is required to run nested **ESXi hosts and the vCenter Server Appliance (VCSA)**.

Risks Introduced

Disabling these protections reduces the defense-in-depth posture of the system and introduces the following risks.

- **Reduced firmware protection** due to modifications of System Management Mode (SMM) safeguards
- **Reduced kernel memory isolation** resulting from disabling Virtualization-Based Security (VBS)
- **Increased credential exposure risk** when Local Security Authority (LSA) protection is disabled

Justification

These configuration changes are required to support **enterprise-grade nested virtualization labs**, specifically environments running

- **VMware ESXi**
- **VMware vCenter Server Appliance (VCSA)**

Nested virtualization allows the simulation of **enterprise infrastructure environments** for testing, training, and cybersecurity research.

Mitigation

All security configurations were **restored to their original state after lab validation**.

Restoration included:

- Re-enabling **Virtualization-Based Security (VBS)**
- Restoring **Local Security Authority (LSA) protection**
- Reapplying default **firmware security settings**

This ensured that the host system returned to its **standard security posture** after the virtualization lab environment was completed.

2.0 CONCLUSION

2.1 Key Takeaways

Hardware Enablement

The Acer Nitro V15 requires a nonstandard firmware startup sequence to access and configure hidden CPU virtualization controls. These settings must be enabled before AMD-V (Server Management Virtualization) extensions become available to the operating system.

Security Conflicts

Windows 11 security features, such as Virtualization-Based Security (VBS) and Hyper-V, load a native Microsoft hypervisor. When active, this layer prevents VMware Workstation from accessing AMD-V/SVM extensions directly. These features must be disabled to support nested virtualization.

Driver Stability

Installing the official AMD Chipset Software package is critical for proper firmware-to-OS communication. The AMD PSP and AMD PMF drivers ensure that Windows correctly detects virtualization capabilities and maintains stability during nested virtualization workloads.

Recovery Readiness

Modifying firmware security features, particularly System Management Mode (SMM) protections, can lead to system instability or a No-POST condition. Firmware recovery procedures, such as a CMOS/NVRAM reset, are essential safeguards when adjusting these settings.

Conclusion on Capacity

The AMD Ryzen 5 6600H platform, combined with a 32GB RAM configuration, provided sufficient resources to support a nested **vSphere 8.0 environment**. This configuration mitigated common nested virtualization constraints, such as CPU contention and memory ballooning, allowing the **vCenter Server Appliance (VCSA)** to operate reliably alongside nested ESXi hosts within VMware Workstation.

2.2 Security Implications and Recommendations

The configuration steps described in this guide require temporarily disabling several built-in Windows and firmware security protections. Although necessary to support nested virtualization testing, these changes reduce the default security posture of the host system.

Security Risks

Disabling these protections introduces the following risks.

- A disabled **Virtualization-Based Security (VBS)** increases exposure to kernel-level attacks by removing virtualization-backed memory protections.
- A disabled **Local Security Authority (LSA) Isolation** increases the risk of credential dumping and privilege escalation attacks.
- **System Management Mode (SMM) Protections Disabled** weakens firmware-level protections designed to isolate privileged system management operations.

Technical Recommendations

After completing the virtualization of the laboratory environment, all security settings should be restored.

UEFI Security Settings

Setting	Recommended State
SMM Code Lock	Enabled
SMM Isolation	Enabled
SVM Lock	Enabled if nested virtualization is no longer required

Restore Windows Security Features

Re-enable Windows hypervisor

```powershell
bcdedit /set hypervisorlaunchtype auto
```

Re-enabling Virtualization-Based Security

```powershell
reg add "HKLM\SYSTEM\CurrentControlSet\Control\DeviceGuard"
/v "EnableVirtualizationBasedSecurity" /t REG_DWORD /d 1 /f
```

Re-enable Local Security Authority protection

```powershell
reg add "HKLM\SYSTEM\CurrentControlSet\Control\LSA" /v "LsaCfgFlags"/t
REG_DWORD /d 1 /f
```

After these changes, a full system restart is required.

Procedural Recommendations

- Consider these configuration changes as **controlled exceptions in the laboratory**.
- These settings should not be applied **to production systems** without formal approval.
- Maintain documentation of all firmware and operating system modifications.

Framework Alignment

These practices align with the established cybersecurity configuration management standards.

NIST SP 800-53

- **CM-2 – Baseline Configuration**
- **CM-6 – Configuration Settings**
- **SI-2 – Flaw Remediation**

Configuration changes should be **controlled, documented, and reversible**.

CIS Benchmarks (Windows 11) Disabling VBS and LSA protections deviates from the recommended baseline and should only be performed temporarily in controlled environments.

The Least Functionality Principle disables only the features required to complete the lab task and restores them immediately afterward.
